

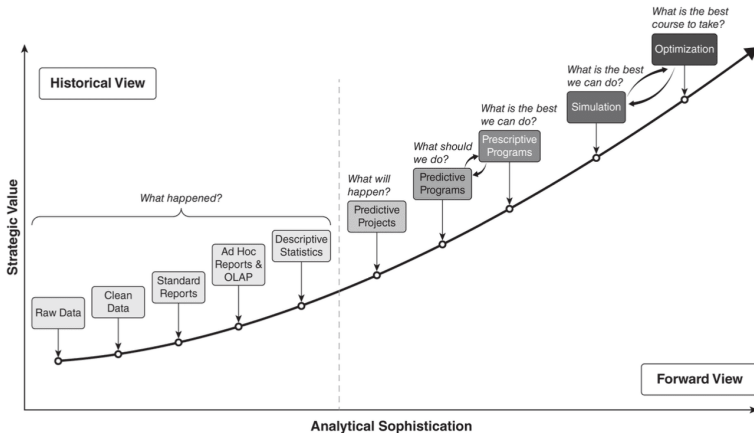
Big Data and Traditional AI

- **Focus for past 10 years:**
 - Collect and organize massive data volumes
 - Run analytics (dashboards, models, reports) on data
- **What Are Data Analytics?**

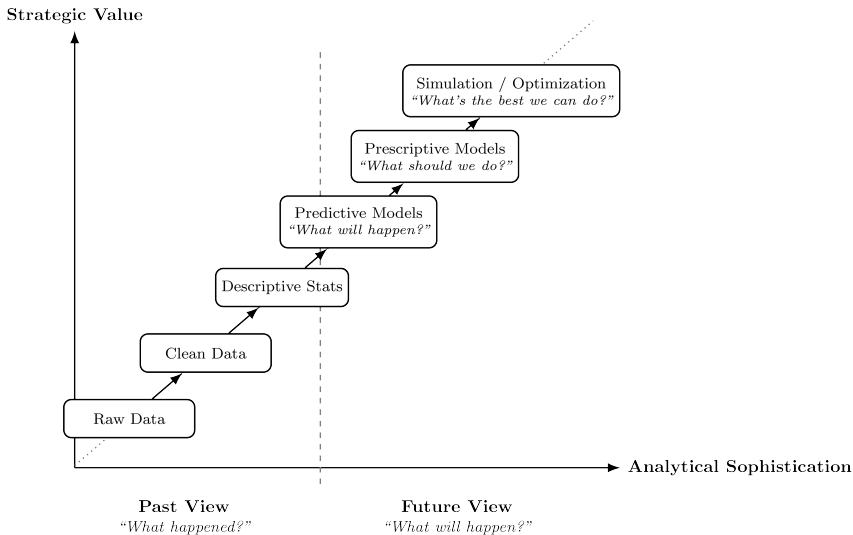
Category	Description	Example
Collections of data	Organized datasets for analysis	Customer purchase histories in CRM
Descriptive statistics	Key metrics: mean, median, standard deviation	Average quarterly sales to track trends
Historical reports	Review of past performance	Monthly sales reports from last year
Dashboards	Visuals of key metrics for quick insights	Quarterly revenue and expense dashboard
Models	Statistical tools to forecast, explain	Churn prediction from customer behavior

Data Analytics Sophistication

Business Question	Methodology
What happened?	Descriptive statistics
What will happen?	Predictive models
What should we do?	Prescriptive programs
What's the best we can do?	Simulation + optimization



Test



- *What ML Systems Can and Cannot Tell You*
- Why Causal AI Matters

The Illusion of Understanding

- **Prediction vs. Explanation**
 - Models find patterns in historical data
 - Can't explain mechanisms or answer "what if" scenarios
- **High accuracy $\neq$ Understanding causation**
 - A model with 95% churn prediction accuracy won't tell you what *causes* churn
 - Accurate sales forecasts can't predict effects of *pricing changes*
- **The Real Problem**
 - Without causation, you react to past events
 - You can't proactively *change outcomes*

Association, Correlation, and Causation

- **Association**

- Variables co-occur or are statistically dependent
- E.g., “*People who buy diapers also buy beer*”
- No info on direction or mechanism

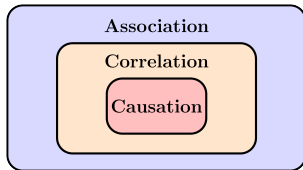
- **Correlation**

- Linear relationship between variables
- E.g., “*Ice cream sales and drowning deaths correlate*” (both seasonal)

- **Causation**

- Changing one variable *directly* changes another
- E.g., “*Asbestos causes cancer*”
- Requires causal model or experiment

- **The illusion**: high correlation *feels like* understanding
 - Real understanding means knowing what happens when you *intervene*



What ML Systems Can Tell You

- **Pattern recognition**
 - Find correlations in data
 - Identify similar items (e.g., through clustering, classification)
 - Forecast based on past trends
- **Predictive accuracy on held-out test data**
 - Accurate prediction if conditions remain stable
 - Only works when training and future data match
- **Feature importance**
 - Find variables with strongest statistical associations
 - Doesn't distinguish correlation from causation

What ML Systems Cannot Tell You

- **Causation from observational data alone**
 - Strong correlations can arise from confounding, reverse causation, or coincidence
- **Effects of interventions**
 - “If we change X , what happens to Y ?”
 - E.g., “If we lower prices by 10%, will revenue increase?”
 - Observational data can't answer without causal assumptions
- **Counterfactuals**
 - “What would have happened with a different decision?”
 - E.g., “Would customer churn if we offered them a discount?”
- **Fairness**
 - A model can be *statistically* unbiased but *causally* biased
 - *Statistically unbiased*: $\mathbb{E}[\text{Prediction}] = \mathbb{E}[\text{TrueValue}]$
 - *Causally biased*: model uses variables that are proxies for protected attributes, even if those variables aren't explicitly protected
 - E.g., using variables correlated to race, gender in decisions
- **Optimal decisions**
 - ML optimizes for accuracy, not business outcomes
 - E.g., a 90% accurate model might lead to worse decisions than an 85% accurate one, depending on the consequences of

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Problem 1: Correlation is Not Causation!

- **Correlation**: statistical relationships between variables
 - Pros: Find patterns in past data to predict future
 - Cons:
 - Doesn't explain cause
 - Variables move together by coincidence or hidden factors
- **Causation**: explains how changing one variable influences another
 - Cannot be concluded from correlation alone
- **Data doesn't understand causes and effects**
 - Only humans identify variables and relationships from context
 - Without causation, you can't make intelligent decisions

Correlation is Not Causation: Examples

- Intuitively you know that association is not causation
- Humans confuse association with causation
 - “*You eat X, get stomachaches*” → infer “*X is bad for you*”
 - “*Buy stock before price spike*” → infer “*you can time markets*”
 - Leads to riskier bets
- E.g., “top-notch consulting improves businesses”
 - Or do thriving businesses hire top consultants?

Correlation is Not Causation: Examples

- E.g., in hotel industry
 - Prices are low outside tourist season
 - Prices are high when demand is high and hotel full
 - A naive prediction suggests to increase prices to sell more rooms
- E.g., online marketplace company
 - Decide how to set prices and when to have sales
 - What is the impact of lowering prices on units sold?
 - Gains from selling more need to compensate for the loss of selling cheaper
 - This is a causal question: “what would happen if we lower prices vs doing nothing”
 - Price cuts can have different impacts depending on
 - The type of business (e.g., clothing vs dog food)
 - The timing (e.g., before Christmas)